KCC Clean City App — Feature Specification

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Overview

The new app gives Khulna City Corporation (KCC) citizens a photo-and-GPS complaint channel for waste, and gives KCC a closed loop from report to verified clean-up. It is built on the existing SWMS Django/DRF backend, so complaints, customers, fleet and billing share one database.

The reference is Clean Care, the DSCC app reviewed from 34 screenshots and the Clean Care feature spec. Clean Care gets the citizen side right: a 2-tap category → sub-issue picker, photo plus voice plus text, ward auto-detected from GPS, a notice board and officer messages. It stops at the citizen: live vehicle tracking and the ward dashboard are advertised but not shown in the app, and there is no collector-side workflow, no proof-of-cleaning photo and no SLA visible to the citizen.

Area	Clean Care (DSCC)	KCC app (this spec)
Complaint wizard	10 categories, 4–7 sub-issues, photo, 2-min voice, text	Same flow, KCC categories, plus duplicate detection and map pin adjust
Status tracking	List only, no timeline seen	Timeline with SLA countdown and before/after photos
Collector / field side	Not in the app	Separate field app: tasks, navigation, proof photos, GPS trail
Vehicle tracking	Advertised, not shown	Live truck on map and ETA for the citizen's road
Pickup schedule	Government holiday calendar only	Ward-wise collection calendar with reminders
Billing	"Billing issue" as a complaint type only	View and pay household waste bill (bKash / Nagad)
Language	Bangla + English	Bangla + English, Bangla first

Target users are KCC households and businesses in all 31 wards, ward conservancy staff, PCSP/contractor crews and KCC conservancy department officers.

User roles and apps

The system is three front ends on one SWMS backend: a citizen app, a field app for staff, and the KCC web admin panel.

Role	Front end	Main jobs	Data scope
Citizen (household / business)	Citizen mobile app (Android first, iOS Phase 3)	Report, track, rate, view schedule, pay bill, read notices	Own complaints and bills
Collector / crew (PCSP or contractor)	Field app (Android)	Receive tasks, navigate, upload before/after photos, mark done, daily route check-in	Tasks assigned to them
Ward supervisor / conservancy inspector	Field app + admin panel	Assign and re-assign, verify clean-up, handle escalations	One or more wards
Zone / conservancy officer	Admin panel	Monitor SLA, approve closures, publish ward notices	Zone
KCC admin / Chief Conservancy Officer	Admin panel	Categories, SLA rules, users, notices, reports, officer messages	All wards
Support agent	Admin panel (chat console)	Answer chat and hotline, create complaints on behalf of callers	All complaints, read + comment
System admin	Django admin	Configuration, integrations, audit	Everything

Citizen app modules and screens

The citizen app keeps Clean Care's bottom bar (Home · red camera FAB · Complaints) and adds a Schedule tab and a Bills tab, so the five things citizens do most are one tap away.

Bottom navigation: Home · Schedule · Camera FAB (new complaint) · My Complaints · Profile. Header: logo, "KCC Clean City" + tagline, notification bell with badge, hamburger menu.

#	Screen	Elements	Notes vs Clean Care
1	Splash + onboarding	3 slides (report, track, schedule), language pick (বাংলা / English), permission primer	New
2	Register / login	Phone + OTP (primary), optional email + password; name, ward, address; Google sign-in optional	Clean Care uses phone + email + password
3	Home	Officer message carousel (Mayor / CEO / Chief Conservancy Officer, photo + quote); 4 quick tiles around a central Complaint button: Customer Care, Live Chat, Waste Management, Emergency; "Next pickup: Tomorrow 7–9 AM" card; live truck card ("Truck 0.8 km away, ETA 12 min"); KCC Notice Board ticker; "Visit kcc.gov.bd" banner; first-run coach-mark tooltip on Complaint	Pickup and truck cards are new
4	Camera confirm	"Use the camera to file a waste complaint?" Yes/No; full-screen camera with flash toggle; Use Photo / Retake	Same
5	Choose category	2-column colour-coded tiles, Bangla title + English subtitle; hint text at the bottom	Same pattern, KCC categories
6	Select problem type	2-column sub-issue cards with icon; every category ends with "অন্যান্য / Others"	Same
7	Complaint details	Selected category + sub-issue card; up to 4 photos; voice note up to 2 min; description (Bangla/English); Continue to Address	Max 4 photos added

#	Screen	Elements	Notes vs Clean Care
8	Address & location	Map with draggable pin (GPS auto-fill); auto ward + zone + area via ward polygons; house/road/landmark fields; "Use my saved address"	Clean Care shows this step only in the spec
9	Review & submit	Summary, duplicate warning ("2 open reports within 50 m — join instead?"), submit; success screen with tracking ID (e.g. KCC-W12-000231)	Duplicate check new
10	My Complaints	Filter chips (All, Open, In progress, Resolved, Closed); card with photo, category, ward, status pill, age; empty state "No complaints yet" + Submit Complaint	Same + filters
11	Complaint detail	Status timeline; assigned crew name + phone mask; SLA countdown; before/after photos; complaint chat thread; Reopen (within 48 h); rate 1–5 stars + comment	Timeline, proof and SLA new
12	Schedule	Ward collection calendar (green = collection day, red = public holiday); time window; reminder toggle; Government Calendar 2026 image as a sub-tab	Clean Care shows only the holiday image
13	Live tracking	Map of trucks on the user's route, ETA to saved address, last-seen time	Advertised by Clean Care, not built
14	Bills	Current holding waste bill, due date, history, pay via bKash / Nagad / card, receipt PDF	New
15	Notice Board	Filter chips (All, Campaign, Emergency, Event, Holiday); timeline cards with URGENT / category badges, date, views, like and heart counts	Same

#	Screen	Elements	Notes vs Clean Care
16	Notifications sheet	Bottom sheet list, Mark all read, empty state	Same
17	Photo Gallery	KCC clean-up activity albums	Same
18	Officer Reviews	List of officers (photo, title, comment count) → detail with message and citizen comments	Same
19	Customer Care / Live Chat	Hotline call button, WhatsApp link, email; live chat with text, photo, voice	Same
20	Emergency	Report urgent hazards (dead animal, chemical/medical waste, fire in dump, blocked drain flooding) with 2-hour SLA; emergency numbers list	Expanded
21	Waste Management guide	Segregation guide (wet / dry / hazardous), do's and don'ts, recycling points map, bulk pickup request	New content
22	Profile	Avatar initials, name, phone, Edit Profile; account info (email, phone, role, status); Administrative Area (City Corporation, Zone, Ward, Address); saved addresses	Same + multiple addresses
23	Settings	Language EN/বাং, push and email toggles, pickup reminder time, dark mode	Same + reminder
24	About	Mission, KCC authorization ref no., features grid, How to Use accordions (Track Status, Rate Resolution, Contact Support), contact list, KCC website, Privacy Policy, version	Same
25	Delete account	Red button → reason dropdown → confirm with OTP	Same + OTP

Hamburger menu: Profile Settings, My Complaints, Government Calendar, Notice Board, Photo Gallery, Officer Reviews, About, Language switch, Logout.

Complaint category tree

The 10 categories and their sub-issues below are taken from the Clean Care screens, with KCC additions marked (new). They live in the `catalog` app as admin-editable data, so KCC can rename or add items without an app release.

Category (Bangla / English)	Sub-issues	Default SLA	Routed to
বাসা/বাড়ি — Home/House	Household waste not collected; Behaviour of waste collectors; Billing issue; Irregular pickup time (new); Others	24 h	Ward PCSP / contractor
রাস্তা ও নর্দমা — Road & Drainage	Waste beside the road; Water logging on road; Drainage issue; Missing manhole cover; Others	12 h (manhole: 4 h)	Ward conservancy inspector
ব্যবসা প্রতিষ্ঠান — Business Place	Waste not collected; Behaviour of collectors; Billing issue; Others	24 h	Ward PCSP / contractor
অফিস — Office	Waste not collected; Behaviour of collectors; Billing issue; Others	24 h	Ward PCSP / contractor
শিক্ষা প্রতিষ্ঠান — Educational Institution	Waste not collected; Behaviour of collectors; Billing issue; Others	24 h	Ward PCSP / contractor
হাসপাতাল — Hospital	Waste not collected; Behaviour of collectors; Billing issue; Medical waste mixed with general waste (new); Others	12 h	Conservancy + health dept.
ধর্মীয় ও সেবামূলক — Religious & Service	Waste not collected; Behaviour of collectors; Billing issue; Others	24 h	Ward PCSP / contractor
মেলা ও আনন্দোৎসবের সৃষ্ট ময়লা — Events & Celebration Waste	Fair waste; Celebration waste; Event waste; Eid-ul-Azha animal waste (new, seasonal); Others	12 h	Zone conservancy officer

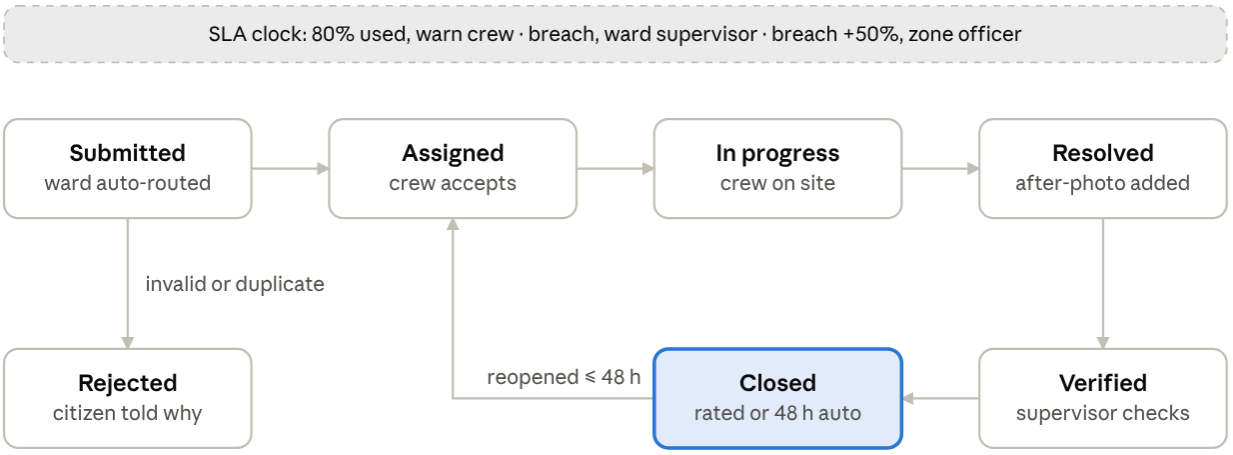
Category (Bangla / English)	Sub-issues	Default SLA	Routed to
খাল ও জলাশয় — Canal & Water Body	Waste accumulated in canal; Waste in water body; Canal is blocked; Water pollution; Others	48 h	Canal / drainage section
নর্দমা ও জলাবদ্ধতা — Drainage & Waterlogging	Drainage blocked; Waterlogging; Missing drainage cover; Bad smell; Others	12 h	Drainage section
জরুরি — Emergency (new)	Dead animal; Hazardous / chemical waste; Fire at dump or bin; Flood-blocked drain	2 h	Duty officer, 24/7
স্বাস্থ্য উন্মুক্ত ময়লা — Illegal Dumping (new)	Open dumping spot; Waste burning; Construction debris; Others	24 h	Ward conservancy inspector

Each category record stores: Bangla and English names, icon, colour, sort order, default SLA hours, routing role, whether a photo is mandatory, and active flag. SLA figures are proposals for KCC to confirm.

Complaint lifecycle, SLA and escalation

A complaint moves through seven states; the citizen sees each change as a timeline step and a push, and the SLA clock runs from submission to resolution.

A complaint closes only after a supervisor verifies the after-photo



complaint lifecycle · 7 states

If the supervisor finds the spot still dirty, Verified sends it back to In progress. A citizen can reopen a closed complaint within 48 hours; the second reopen goes straight to the zone officer. Status changes are written to `ComplaintStatusLog` with actor, time, GPS and note, so every step is auditable.

Features beyond Clean Care

These features close the gaps found in Clean Care and use modules SWMS already has (fleet, billing, fieldops, ai).

Field app for collectors and supervisors (Android, offline-first)

- Daily duty check-in with selfie + GPS; route of the day with assigned holdings and bins.
- Task inbox: new complaints pushed by ward, with map, photos, voice note and citizen's masked phone.
- Accept → On the way → Arrived (geofence 50 m) → Upload after-photo → Done. Before/after photos must be taken in-app with GPS and timestamp watermark.
- Can't-complete reasons: wrong location, needs heavy equipment, private property, other — sends back to supervisor.
- Background GPS trail every 30 s while on duty; works offline and syncs when the network returns.
- Supervisor mode: verify closures, re-assign, see crew on map.

Live vehicle tracking and ETA — trucks report position from a GPS device or the driver's field app; citizens see the trucks serving their ward and an ETA to their saved address; "Truck is near" push at 500 m.

Ward collection schedule — per-ward days and time windows, holiday overrides from the government calendar, reminder push the evening before.

Bill payment — shows the holding's monthly waste-collection fee from `billing`; pay with bKash, Nagad or card; receipt PDF; unpaid reminder.

Duplicate and spam control — open complaints of the same sub-issue within 50 m are offered to "join" (adds a +1 instead of a new ticket); daily limit of 10 complaints per account; report-abuse button; photos screened by `swms.ai`.

AI assist (Phase 4) — suggest category from the photo; estimate pile size (small / medium / truckload); flag photos with no waste.

Bulk and special pickup — book construction debris, garden waste or event waste pickup with a fee quote.

Citizen engagement — rate each resolution; officer reviews with comments; Clean Ward leaderboard; optional reward points for verified reports.

Offline and low-bandwidth — drafts saved on device; photos compressed to about 300 KB; queue retries submission when online; cached notices, complaint list and chat.

Accessibility — Bangla-first UI, voice-note complaints for users who don't type, large touch targets, works on Android 8+.

Admin web panel and dashboards

The admin panel is the existing SWMS Vite SPA extended with complaint operations screens; every role sees only its wards.

Module	Key functions
Complaint queue	List + map view; filters by ward, zone, category, status, SLA state, date; bulk assign; detail with photos, voice, chat, timeline, audit log
Assignment & dispatch	Auto-assign by ward + category routing rule; manual re-assign; crew availability; nearest crew suggestion
SLA & escalation rules	Hours per category / sub-issue; escalation chain per level; business hours vs 24/7; holiday calendar
Live operations map	Trucks, crews, open complaints as pins coloured by SLA state, complaint heatmap

Module	Key functions
Catalog	Categories, sub-issues, icons, colours, SLA, routing, Bangla/English text
Notice board	Create notices with type (Campaign, Emergency, Event, Holiday), urgency, ward targeting, image, schedule, push on publish; view and like counts
Officer messages & reviews	Home carousel messages; moderate citizen comments
Gallery	Albums and photos
Collection schedule	Ward × day × time window; holiday overrides
Users & roles	Citizens (block/unblock), staff, PCSP/contractors, ward mapping
Customer care console	Live chat queue, canned replies, create complaint for walk-in or phone caller
Reports	Complaints by ward/category/status; average and P90 resolution time; SLA breach list; crew performance; citizen rating; contractor scorecard; export to Excel/PDF
Public dashboard	Live counters for the website: total reports, resolved %, average resolution hours, active wards
Settings & audit	App version gate, maintenance banner, audit log, account deletion requests

Key dashboard KPIs: complaints received today, open, overdue, resolved within SLA %, average resolution hours, reopen rate, average rating, and ward ranking.

Mapping to SWMS apps, data model and API

Most of the backend already exists as `swms.*` apps; the new work is concentrated in `complaints`, `fieldops`, a new `content` app for notices and gallery, and a `notifications` app.

SWMS app	Used for	New models / changes
accounts	Citizen and staff users, OTP login, roles	`OTPCode`, `DeviceToken` (FCM), `DeletionRequest`; role field values: citizen, crew, supervisor, officer, admin, support
agencies	City corporation, zone, ward, PCSP/contractor	`Ward.boundary` (PostGIS polygon) for GPS → ward lookup; `Contractor` ↔ ward mapping
catalog	Complaint categories and sub-issues	`ComplaintCategory`, `ComplaintSubCategory` (bn/en name, icon, colour, sla_hours, route_role, photo_required)
customers	Holdings / households, saved addresses	`CitizenAddress` (label, point, ward, holding link)
complaints	Core complaint lifecycle	`Complaint`, `ComplaintMedia` (photo/voice, before/after), `ComplaintStatusLog`, `ComplaintAssignment`, `ComplaintMessage`, `ComplaintRating`, `ComplaintSupport` (+1 joins), `SLAPolicy`, `EscalationEvent`
fieldops	Crew duty, tasks, GPS trail	`DutyShift`, `CrewLocation` (time-series), task view over `ComplaintAssignment`
fleet	Trucks, routes, live position	`VehiclePosition`, `RouteStop` for ETA
billing	Waste fee per holding, payments	`PaymentTransaction` (gateway, ref, status), bKash/Nagad callbacks
surveys	Satisfaction surveys	Post-resolution rating can reuse survey answers
reports	Dashboards, exports	SLA, ward ranking, contractor scorecard queries
ai	Photo classification	Category suggestion, no-waste flag

SWMS app	Used for	New models / changes
content (new)	Notice board, officer messages, gallery, schedule	`Notice`, `NoticeReaction`, `OfficerMessage`, `OfficerComment`, `GalleryAlbum`, `GalleryPhoto`, `CollectionSchedule`, `Holiday`
notifications (new)	Push, SMS, in-app inbox	`Notification`, `NotificationPreference`; Celery tasks for FCM and SMS

Core `Complaint` fields: `tracking_no` (KCC-W{ward}-{seq}), `citizen`, `category`, `sub_category`, `description`, `location` (Point), `address_text`, `ward`, `zone`, `status`, `priority`, `assigned_to`, `sla_due_at`, `resolved_at`, `closed_at`, `reopen_count`, `support_count`, `source` (app, web, hotline), `created_at`.

Citizen API (`/api/v1/`, drf-spectacular documented):

Method	Endpoint	Purpose
POST	`/auth/otp/request/`, `/auth/otp/verify/`	Phone login
POST	`/auth/token/refresh/`	Refresh JWT
GET / PATCH	`/me/`, `/me/addresses/`	Profile and saved addresses
GET	`/catalog/categories/`	Category tree with sub-issues
GET	`/geo/locate/?lat=&lng=`	Ward, zone and area from GPS
GET	`/complaints/nearby/?lat=&lng=&sub=`	Duplicate check
POST	`/complaints/` (multipart)	Create with photos and voice
GET	`/complaints/mine/`, `/complaints/{id}/`	List and detail with timeline
POST	`/complaints/{id}/support/`, `/reopen/`, `/rate/`	Join, reopen, rate
GET / POST	`/complaints/{id}/messages/`	Complaint chat
GET	`/schedule/?ward=`, `/fleet/vehicles/live/?ward=`	Pickup schedule, live trucks + ETA
GET / POST	`/billing/invoices/`, `/billing/pay/`	Bills and payment

Method	Endpoint	Purpose
GET	`/notices/`, `/gallery/`, `/officers/`	Content
GET / POST	`/notifications/`, `/notifications/read-all/`, `/devices/`	Inbox and FCM token
POST	`/me/delete/`	Account deletion

Field API: `/field/shift/start|end/`, `/field/tasks/`, `/field/tasks/{id}/accept|arrive|resolve|reject/`, `/field/location/` (batched GPS points).

Mobile stack suggestion: React Native (Expo) for both citizen and field apps, sharing one TypeScript API client generated from the OpenAPI schema; WebSocket (Django Channels) for chat and live truck positions; Celery + Redis for SLA timers and push.

Notifications, security, privacy and account deletion

Notifications go by FCM push first, with SMS only for OTP and for emergency complaints.

Event	Citizen	Crew	Supervisor
Complaint submitted	Push + tracking ID	—	Push (new in ward)
Assigned / accepted	Push	Push (new task)	—
Crew on the way / arrived	Push	—	—
Resolved (with after-photo)	Push + rate prompt	—	Push (verify)
SLA 80% used / breached	—	Push	Push, then escalate
Reopened	—	Push	Push
Truck within 500 m	Push (if enabled)	—	—
Pickup reminder	Push evening before	—	—
New notice / emergency alert	Push (ward-targeted)	Push	Push
Bill due / paid	Push + receipt	—	—

Security: JWT access tokens (15 min) with rotating refresh tokens; OTP rate limiting; HTTPS only; passwords hashed with Argon2 or PBKDF2; object-level permissions by ward; media in private storage with signed URLs; citizen phone numbers masked to crews; full audit log of status changes.

Permissions requested: camera, photos/storage, microphone (voice notes), location (while in use; background only in the field app), notifications.

Privacy: data shared only with KCC and the ward PCSP/contractor; complaint photos shown publicly only after face and number-plate blurring (Phase 4); privacy policy and terms linked in About and at registration, written to Bangladesh's personal data protection rules.

Account deletion, in the app (Profile → Delete Account → reason → OTP) and on a web page for Google Play compliance:

Data	Treatment
Name, phone, email, photo, addresses	Deleted
Password hash, sessions, refresh tokens, FCM tokens	Deleted immediately
Chat, notification history	Deleted
Past complaints and their photos	Kept anonymised for KCC audit
Payment records	Kept as required for accounts, unlinked from the profile
Backups	Purged within 30 days

Release phases and open questions

Phase 1 delivers the full complaint loop; tracking, billing and analytics follow once complaint data is flowing.

Phase	Duration	Scope	Exit criteria
1 — Complaint MVP	8 weeks	Citizen: register/OTP, home, complaint wizard, my complaints, notices, profile, delete account. Field: task list, accept, proof photo. Admin: queue, assign, notice board.	3 pilot wards live; 90% of complaints assigned within 1 h

Phase	Duration	Scope	Exit criteria
2 — Operations	6 weeks	Collection calendar, live truck tracking + ETA, field GPS trail, SLA escalation engine, complaint chat, ratings, officer messages	All 31 wards; SLA breach report used weekly
3 — Revenue & insight	6 weeks	Bill view + bKash/Nagad payment, ward dashboard, public impact counters, heatmaps, emergency/bulk pickup booking, iOS release	Bill payments reconciled daily; public dashboard live
4 — Smart city	Ongoing	AI auto-category from photo (swms.ai), duplicate detection by image, bin-fill IoT sensors, citizen reward points	Decided after Phase 3 review

Open questions:

- [ ] Does KCC want customer support as in-app live chat, a hotline, or both? Who staffs it?
- [ ] Which payment gateway is approved by KCC finance: bKash, Nagad, SSLCommerz, or all?
- [ ] Are PCSP/contractor crews KCC employees or vendor staff, and who issues their phones?
- [ ] Which 3 wards pilot Phase 1?
- [ ] Is the app name "KCC Clean City" final, and is KCC's official approval reference available for the About screen?
- [ ] Do trucks already carry GPS trackers, or does the field app act as the tracker?
- [ ] SLA hours per category need sign-off from the Conservancy department.